# Vibe Coding For AI Application Development

Teaching Record

24-07-2026

# Scope

- Vibe Coding
  - What/Why/How
  - Tools
  - Python
  - IT Essentials
- AI Application Development
  - Workflow
  - Chatbot
  - Agents

# Scope

- Overview of AI Applications with Vibe Coding
- Setup of tools
- Python with vibe coding
- Claude Code
- Use cases of AI applications
- Agentic use cases with vibe coding

# Use cases of AI Solution Development

- Document Summarization Engine
  - OCR detection engine
- Database Design and deployment
  - Text2SQL
- Web scraping techniques
  - News sentiment Analysis engine
- Multi-Agents deployment

# Class One

- Overview of LLM Solutions
  - Prompts
  - Coding LLM
  - Agent
    - What is Agent
      - Execution
      - LLM (Brain)
      - Tools (Execution)
      - Memory
      - Harness (looping and orchestration)
- Setup of tools
  - Cursor
    - Subscription
  - Grok/Gemini (Browser mode LLM)
  - Claude code CLI (To be continued)
    - Cursor
    - Openrouter (LLM)
- Use sample py program to read it with coding agent

# Class one

- Use case of using cursor
  - Read existing python script
  - Execute the python script
- Prompt techniques (To be continued)
- Python programming with vibe coding

# Class two

- Recap of last lesson
  - **Grok (Browser)**
  - **Cursor**
    - **Write the program(s) in the working folder**
    - **Debug the program(s) in the working folder**
    - **Deploy the solution in cloud**
  - Claude code
- Prompt techniques
  - System prompt
  - User prompt
  - Zero shot
  - Few shot
- Python programming with vibe coding
  - Tools (Anaconda) (Jupyter notebook)
  - Data types
  - Function
  - **Data structures (To be continued)**
- **Class exercise**
  - **Vibe coding**
    - **Generate stock processing excel**

# Class two

- Function
  - From python native functions
    - print()
  - From libraries
    - yfinance
      - download()
    - yf.download()
  - Define custom function
- Packages
  - pip install
- Python
  - Data types

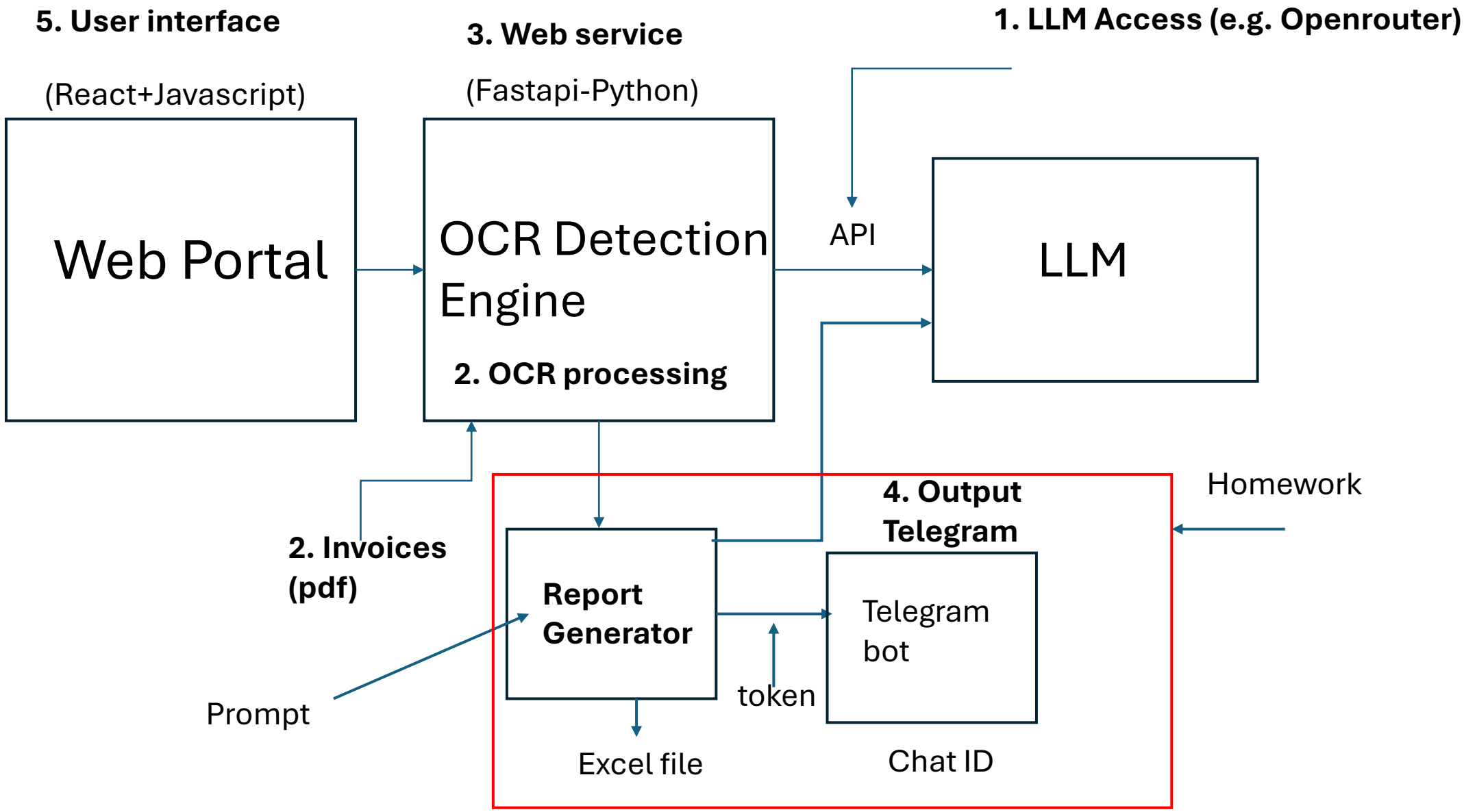
# Class three

- Data objects
  - List
    - append()
  - Dictionary
  - Pandas
    - Series
    - Dataframe
      - describe()
      - Add column
      - Filtering by position
- Homework

# Class four

- LLM API
- OCR Detection engine (Base module)
- Class exercises (Vibe Coding)
- Task automation with Python
- Development of SDD
- Development of OCR detection engine (Front end + Backend)
- Telegram API
- Class exercises

# Home work


5. User interface
(React+Javascript)
3. Web service
(Fastapi-Python)
1. LLM Access (e.g. Openrouter)
Web Portal
OCR Detection Engine
2. OCR processing
API
LLM
2. Invoices (pdf)
4. Output Telegram
Homework
Report Generator
Telegram bot
Prompt
token
Excel file
Chat ID

# Class five

- Homework
- Telegram API
- n8n
- MCP
- Fastapi (Python) (To be continued)

# Homework one

- Using the OCR module, add the telegram module and send out message.
- In the front end , when user click the button, it will send the excel file via telegram

# Homework two

- Modify the existing n8n workflow (Personal assistant v3 (6-Aug-2026).json) with the following features:
- UI: Telegram trigger
- AI Agent: Revise the system prompt
- LLM: via openrouter

- Use case: This is a standard AI chatbot and it will reply to user’s question.